

Pénélope Azuelos

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Education

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| 2022 – present | University of Bristol, UK
<i>PhD in Mathematics (ongoing), supervised by Mark Hagen and Indira Chatterji</i> |
| 2021 – 2022 | École Normale Supérieure de Lyon, France
<i>Master (M2) Mathématiques Avancées</i> |
| 2017 – 2021 | University of Bristol, UK
<i>Master in Science in Mathematics (with first class honours)</i> |

Publications

- (with Damien Gaboriau) *Perfect kernel and dynamics: from Bass-Serre theory to hyperbolic groups*, *Mathematische Annalen* 391: 4733–4789, 2025.
<https://doi.org/10.1007/s00208-024-03038-w>
- On subgroups with narrow Schreier graphs*, *Bulletin of the London Mathematical Society*, 56(12): 3652–3668, 2024. <https://doi.org/10.1112/blms.13157>

Preprints

- Uncountably many homogeneous real trees with the same valence*, arXiv:2511.03722.
- A guide to constructing free transitive actions on median spaces*, arXiv:2507.22230.

Teaching

- Teaching assistant at the University of Bristol for the following courses:*
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| Spring 2024 | Introduction to proofs and group theory MATH10010 |
| Autumn 2023 | Combinatorics MATH30030 |
| Spring 2020 | Introduction to geometry MATH20004 |

Conference Talks

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| November 2025 | Parole aux jeunes scientifiques du RT tétraèdre de Platon, Montpellier, “Group structures on real trees and their products”. <ul style="list-style-type: none">– Operator algebras in the south of England, University of Bristol, “Narrow graphs and virtual fiber subgroups”. |
| June 2025 | Conference on hyperbolicity and generalizations, HIT, Harbin, “A guide to constructing free transitive actions on median spaces”. |
| January 2025 | RSME’s 7th congress of young researchers, parallel session on geometric group theory, UPV/EHU, Bilbao, “Narrow graphs and virtual fiber subgroups” |

September 2024	World of Geometric Group Theory (online), <i>“A geometric criterion for detecting virtual fiber subgroups”</i> .
March 2024	Rigidité en théorie des groupes : aspects géométriques et ergodiques, Université Paris-Saclay, <i>“Un critère géométrique pour déterminer si un sous-groupe est une fibre virtuelle”</i> .
May 2023	Measured Group Theory, Stochastic Processes on Groups and Borel Combinatorics, CIRM Marseille, <i>“Perfect kernels and dynamics: from Bass-Serre theory to hyperbolic groups”</i> .

Seminar Talks

October 2025	Bristol Junior Algebra and Combinatorics Seminar, <i>“From groups to median algebras and back again”</i> .
September 2025	Séminaire Groupes et Actions, Université Paris-Saclay, Orsay, <i>“Group structures on real trees and their products”</i> .
February 2025	Bristol Junior Geometry Seminar, <i>“Real trees and the groups that act on them”</i> .
December 2024	Séminaire d’Algèbres d’Opérateurs, IMJ-PRG, Paris, <i>“Comment repérer des fibres virtuelles”</i> .
October 2024	Séminaire de l’équipe d’algèbre, topologie et géométrie, Laboratoire J.A. Dieudonné, Nice, <i>“Comment trouver des fibres virtuelles”</i> .
September 2024	Séminaire de Théorie Ergodique, Institut de recherche mathématique de Rennes, <i>“L’espace des sous-groupes : noyau parfait, transitivité topologique et actions moyennables.”</i> – Algebra Geometry and Topology seminar, Heriot-Watt, <i>“Narrow graphs and virtual fiber subgroups”</i> .
May 2024	International Young Seminar on Bounded Cohomology and Simplicial Volume, <i>“Topology and dynamics on the space of subgroups”</i> .
March 2024	GdT Action ! ENS de Lyon, <i>“Un critère géométrique pour déterminer si un sous-groupe est une fibre virtuelle”</i> .
February 2024	Cambridge Junior Geometry Seminar, <i>“Ends of groups and related invariants”</i> .
December 2023	Bristol Postgraduate Seminar, <i>“Is it trivial? And other difficult problems”</i> .
November 2023	Oxford Junior Topology and Group Theory Seminar, <i>“Topology and dynamics on the space of subgroups”</i> .
September 2023	Le séminaire virtuel francophone Groupes et Géométrie, <i>“Topologie et dynamique de l’espace des sous-groupes”</i> .
January 2023	Bristol Junior Geometry Seminar, <i>“Subgroup spaces and actions on trees”</i> .

Service

2025	Organisation of the Postgraduate Group Theory Conference 2025
2023 – 2024	Organisation of the Bristol Junior Geometry Seminar

Awards and funding

2024	Grant from the Académie d'Excellence "Systèmes Complexes" for a three month research visit to the LJAD in Nice, supervised by Indira Chatterji.
2022 – 2026	University of Bristol PhD scholarship.
2021 – 2022	MiLyon scholarship for an M2 masters program.
2020	Summer research bursary funded by the Heilbronn Institute for Mathematical Research, supervised by Misha Rudnev.

Languages

English and French (native speaker)